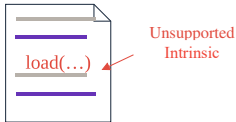


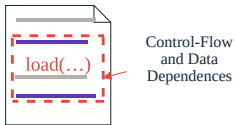
(a) Source Analysis

Locate Ininsics



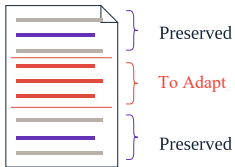
Control-Flow and Data-Flow Analysis

Bound Scope

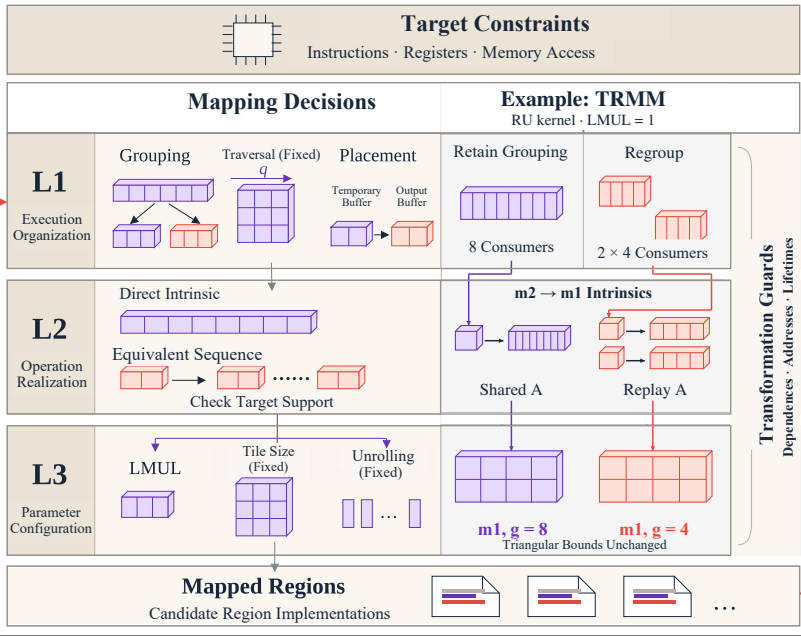


Dependence Closure

Partition Regions

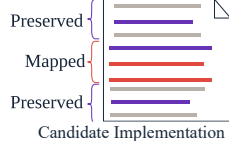


(b) Hierarchical Mapping



(c) Adaptive Selection

Assemble Kernel



Target Compiler



Validate & Measure

- ✓ ISA Checks
- ✓ Numerical Checks
- ⌚ Latency

Autotuning

Kernel Selection



Preserved Structure



Adapted Structure